CHAPTER II

Review of Related Literature/Studies

This chapter presents a brief review of related literature, studies, and interrelated systems that serve as the proponents' basis for developing the Integrated Housing Services and Monitoring System for Local Government Human Settlements.

Foreign Related Systems

Revolutionizing Property Management: e-Perumahan Web-Based Housing Management System

The study by Uriawan, W., Noorsyahbannie, M. N. B., et al. (2023) aimed to develop a web-based housing management platform that centralizes housing data, automates property management processes, and enhances communication among residents, administrators, and service providers. The objective was to address common problems in traditional property management, such as disorganized record-keeping, delayed maintenance replies, and limited access to information. The researchers used the Agile Scrum methodology for system design and development. Interviews and observations were used to gather requirements, the system was modeled with UML and built in PHP with a MySQL database, and black-box testing was performed to assess usability and functionality. The platform includes features such as resident registration, service request submission, maintenance tracking, complaint handling, and digital document storage, and the prototype was assessed based on predefined usability criteria to measure accessibility, efficiency, and overall performance. The results showed improved ease of use, more accurate and accessible information, better coordination among stakeholders, faster response to maintenance concerns and complaints, reduced paperwork, and increased transparency in housing operations. In conclusion, the study concluded that implementing an integrated web-based housing management system enhances administrative efficiency, centralizes housing information, and strengthens stakeholder communication, implying that digital platforms supporting real-time data access, complaint management, and streamlined coordination are essential for responsive, transparent, and accountable housing governance.

Online Complaint Management System

The study of Bhadouria, L. S., Kumar, et al. (2021), published in the Turkish Journal of Computer and Mathematics Education, proposed Online Complaint Management System to improve the registration, monitoring, and resolution of public complaints. The objective of the study was to develop a web-based platform that allows users to submit complaints to appropriate departments, track the status of their concerns, and ensure timely resolution, thereby reducing time consumption and minimizing corruption associated with manual processes. The researchers employed a system design and development method, gathering requirements through analysis of existing complaint procedures and stakeholder interviews, and designing a web-based prototype that was tested for functionality and usability. The system architecture assigns complaints to designated departmental administrators, with automatic escalation to higher authorities if issues remain unresolved within a specific timeframe. It also generates department reports on open, closed, and pending complaints, along with administrator performance evaluations, potentially encouraging improved service delivery through performance monitoring. Additionally, the platform includes a section highlighting available government services to enhance public awareness. Findings indicated that implementing the system improves coordination, monitoring, and tracking of complaints, reduces communication bottlenecks, and enhances administrative responsiveness. The study concluded that automating complaint workflows and providing real-time visibility into complaint status and departmental performance contribute to more efficient grievance management, increased accountability, and time-saving benefits for both the public and service agencies, supporting the broader adoption of digital technologies to strengthen public service delivery and transparency.

Digital Transformation of Public Services: The Case of the Document Management Application

The study by Szedmák, Borbála, et al. (2025), published in the International Journal of Public Administration, examined the implementation and impact of a nationwide digital transformation initiative in Hungary's municipal public services, focusing on a document management application deployed through a centralized Application Service Provider (ASP) platform across more than 3,000 municipalities. The primary objective was to assess how the system contributes to administrative efficiency, transparency, and service performance, particularly in a region where empirical evidence on digital transformation outcomes is limited. The researchers employed a mixed-methods approach, analyzing longitudinal data on case processing times from 2017 to 2022 and conducting 24 expert interviews alongside reviews of feasibility studies and system evaluation reports. Findings indicated a substantial increase in electronic case handling, faster case closure compared to paper-based processes, and reduced administrative lead times. The system facilitated standardized workflows, optimized internal procedures, improved operational efficiency, and enabled broader process reengineering across municipalities. Results also showed enhanced transparency and comparability of public services, with success dependent on technology adoption, process standardization, and centralized IT management. The study concluded that large-scale digital transformation can generate measurable efficiency gains, improve service delivery, and support agile governance, highlighting the importance of centralized platforms, continuous process optimization, and adaptive administrative systems to maximize the benefits of digital technologies in public sector reform.

An Integrated Complaint Management System Based on Large Language Models: Case Study in the Electric Sector

The study by Sierra, C., Huanca, et al. (2025), published in *Array*, explored the design and evaluation of a complaint management system that integrates large language models (LLMs) to improve complaint handling in the electric utility sector. The primary objective was to overcome the limitations of traditional and rule-based complaint resolution methods, such as manual processing and limited ability to interpret unstructured feedback, by incorporating advanced natural language understanding into an automated system. The researchers employed a system development methodology using a service-oriented architecture, which included a data processing microservice, an AI module powered by LLM-based virtual analysts, and a user-friendly web application interface. The LLM component was responsible for classifying, prioritizing, and assisting in the automated resolution of complaints. Evaluation of the system involved load testing and latency analysis to assess performance, scalability, and robustness under operational conditions. Findings showed that the system significantly improved automation, accuracy, and speed of complaint handling, while enabling semantic understanding of unstructured inputs and automated routing. The study concluded that AI-driven complaint management systems can enhance customer satisfaction, streamline workflows, reduce human intervention, and support more efficient and responsive service delivery in public sector environments.

Complaint Management System and Patient Satisfaction in Grassroots Hospitals

The study by Li, G., Chen, Y., and Lou, X. (2024), published in *Medicine*, examined the role of a structured complaint management system in improving patient satisfaction in grassroots hospitals facing resource constraints and high service demand. The primary objective was to analyze the causes of patient complaints and evaluate how implementing a formal complaint-handling process, including categorization, regulatory framework development, and statistical analysis, affects service quality and satisfaction among patients and their families. The researchers employed quantitative methods, using Pearson chi-square tests, Spearman correlation analysis, and univariate logistic regression to assess relationships between complaint characteristics and satisfaction outcomes. Verified complaints were categorized by department, administrative office, and type, and a comprehensive complaint-handling procedure was implemented with expanded reporting channels via on-site submission, telephone, internet, and SMS, supported by dedicated personnel. Results showed significant associations between complaints from departments and administrative offices and overall patient satisfaction, with system implementation leading to improved service quality, higher patient satisfaction, and a reduction in total complaints. The study concluded that a well-structured complaint management system strengthens patient experience and promotes continuous service improvement, highlighting the importance of systematic procedures, coordinated responses, and responsive feedback mechanisms in fostering patient-centered care and sustainable quality enhancement in primary healthcare institutions.

Foreign Related Literatures

Understanding Local Government Digital Technology Adoption Strategies

The study published in Sustainability by David, Anne, and others (2023) conducted a systematic review to examine how local governments adopt digital technologies, with the primary objective of synthesizing existing research on strategies, frameworks, and practices that enable municipal and local government units to implement digital solutions effectively. The researchers observed that many local government entities continue to face challenges such as fragmented infrastructures, limited technical expertise, insufficient funding, and resistance to organizational change, all of which impede digital transformation efforts. Using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology, the study systematically collected and analyzed literature from peer-reviewed journals, conference proceedings, and global case studies on local government technology adoption. Adoption strategies were categorized into technological, organizational, and environmental dimensions, emphasizing the need to align digital initiatives with institutional goals and community demands. Findings indicated that successful adoption depends on strong leadership and strategic planning, workforce readiness and continuous training, and active stakeholder engagement, including collaboration

with citizens and external partners. The review also highlighted persistent barriers such as limited funding, system fragmentation, and cultural resistance within agencies. The study concluded that integrating digital technology adoption with governance strategies enhances service delivery, transparency, and administrative efficiency, and that holistic approaches considering technological capacity, organizational structure, and external pressures yield better public satisfaction and operational performance. The implications stress the importance of exploring adoption strategies in developing countries, assessing long-term impacts of digital initiatives, and strengthening policy frameworks to support sustainable digital transformation, thereby guiding local governments in leveraging technology for operational improvement, citizen engagement, and evidence-based governance.

Current State and Future Challenges in Building Management: Practitioner Interviews and a Literature Review.

The study published in the Journal of Building Engineering examined building management practices in commercial and institutional settings with the objective of identifying operational barriers and proposing improvements to enhance efficiency and energy performance. The researchers observed that although modern tools such as Building Automation Systems (BASs) and Computerized Maintenance Management Systems (CMMSs) are intended to improve energy efficiency and occupant comfort, their effectiveness is often limited by technical and non-technical challenges, including gaps between data availability and actionable insights, inconsistent professional training, and underutilization of management tools. The methodology combined a comprehensive literature review with interviews involving thirty building management professionals, gathering information about their professional backgrounds, tools used, coordination processes, and interactions with occupants. This mixed-methods approach enabled the researchers to document prevailing workflows and identify systemic issues in building operations and energy management. Findings revealed that despite technological advancements, BASs frequently generate large volumes of data without providing meaningful interpretation, leading to challenges such as inadequate data visualization, limited analytics capability, insufficient training, fragmented staff coordination, and minimal integration of occupant feedback into decision-making processes. Based on these findings, the authors proposed short-term improvements, including enhanced training programs, improved tool usability, and better data visualization techniques, alongside long-term goals such as developing integrated management platforms that consolidate diverse data sources and support predictive analytics for maintenance and energy optimization. The study concluded that

improving building management practices requires not only technological enhancement but also strengthened professional capacity and organizational alignment, emphasizing that addressing knowledge gaps, tool utilization issues, and coordination challenges is essential for achieving operational efficiency and sustainable, energy-efficient building performance.

A Systematic Review of Construction Project Performance Metrics

The study published in the Canadian Journal of Civil Engineering by Alshaikh and El-Asmar (2023) conducted a systematic review of construction project performance metrics with the objective of consolidating existing knowledge and identifying gaps in how project success is evaluated across various contexts. The researchers observed that although traditional indicators such as time, cost, and quality remain fundamental measures of performance, modern construction projects increasingly require broader metrics that encompass safety, sustainability, stakeholder satisfaction, and productivity due to their growing complexity. Employing a systematic review methodology, the study gathered and analyzed literature from peer-reviewed journals, conference proceedings, and case studies published over the past two decades, categorizing performance indicators into financial, time-based, quality-based, and non-financial measures. Findings revealed significant inconsistencies and a lack of standardization in current measurement practices, making benchmarking and cross-project comparison difficult. The results further indicated that relying solely on traditional metrics fails to capture the multidimensional nature of project outcomes, particularly in projects influenced by environmental, social, and technological considerations. The study concluded that developing a standardized and comprehensive framework of performance metrics is essential for accurately assessing construction project success, and that incorporating diverse performance dimensions enables practitioners to better address project complexity, stakeholder expectations, and sustainability goals. These insights contribute to advancing research, informing policy, and strengthening evidence-based decision-making in construction project management.

Exploring the E-Complaint Method: Learning from the Malaysian Polytechnic Institutions

The technical paper by Ismail (2021) examined the effectiveness of the e-complaint method used in maintenance management practices at Malaysian polytechnic institutions, aiming to evaluate how the system supports maintenance operations and to identify gaps that hinder strategic decision-making. The study noted that conventional e-complaint approaches, often relying on fragmented communication

channels and lacking integration with comprehensive maintenance management systems (MMS), are limited in facilitating effective operations and facilities management. Using multiple case analyses of polytechnic facilities management units, the study identified key weaknesses, including poor integration with MMS, inadequate support for defect diagnosis, and insufficient tools for planning and tracking maintenance outcomes, resulting in recurring issues and inefficient resource use. Findings suggest that the conventional e-complaint method fails to maximize operational value due to its inability to systematically link complaints with maintenance and asset data. The study recommends developing more comprehensive MMS tools that incorporate decision-making processes and enable real-time data tracking and analysis to improve operational efficiency, reduce repetitive defects, and enhance the accuracy and reliability of information for facility managers. The paper concluded that advancing e-complaint methods and integrating them into sophisticated digital maintenance management systems can significantly improve maintenance outcomes, stakeholder coordination, and evidence-based management of building facilities.

How Can Online Citizen Complaints Provide Solutions to Refine Environmental Management: A Spatio-Temporal Perspective

The study by Jiao et al. (2024), published in *Information Processing & Management*, investigated how online citizen complaints can inform and improve environmental management, with the primary objective of analyzing the temporal, spatial, and sentiment characteristics of citizen-generated complaints to support more effective governance. The researchers noted that traditional complaint-handling methods often overlook valuable information embedded in complaint texts, such as public sentiment and location-based patterns, which are critical for responsive environmental management. The study analyzed 7,657 environmental complaints submitted via an e-government platform in Guangzhou City, China from 2018 to 2021, applying sentiment analysis, spatial hotspot mapping, and temporal trend analysis to examine variations in concerns over time and across locations. Findings revealed that over 94% of complaints expressed negative sentiment, with distinct temporal patterns for issues such as air, noise, and light pollution, and that complaint hotspots did not always align with areas of strongest negative sentiment, highlighting the need for targeted interventions based on nuanced spatial data. Socioeconomic factors, industrial activities, and land-use patterns were found to influence complaint distribution, while no clear correlation was observed between response efficiency and sentiment intensity, suggesting that current governmental responses may not fully address public emotional concerns. The study concluded that systematically analyzing online citizen complaints using sentiment and spatio-temporal insights can

enhance environmental governance, prioritize interventions, improve public satisfaction, and support evidence-based decision-making, providing valuable guidance for future research and the development of digital tools for sustainable urban environmental management.

Local Related Systems

Information Management System with Project Monitoring for Barangay

The study published in the Journal of Engineering, Architecture, and Informatics (JEAI) by the University of Saint Louis developed a Windows-based information management system integrated with a project monitoring module for barangay use, aiming to automate documentation and record-keeping while enabling officials to efficiently track and monitor the status of projects, programs, and community activities. The researchers identified limitations in existing manual systems as a major barrier to effective barangay governance. Using the Systems Development Life Cycle (SDLC) framework and the Rapid Application Development (RAD) approach, requirements were gathered through interviews and observations of barangay hall operations, and the system was developed with Windows-based programming tools linked to a relational database. Evaluation using the System Usability Scale (SUS) by IT professionals, practitioners, and barangay officials revealed high usability and satisfaction, with mean scores of 4.26 for usefulness, 4.48 for overall satisfaction, and 4.19 for ease of use, indicating strong agreement among respondents regarding the system’s effectiveness. The study concluded that integrating project monitoring into a barangay information management system significantly enhances decision-making capacity, consolidates resident data and documentation, and provides practical value for project tracking. The implications for future research emphasize designing systems that extend beyond record-keeping to actively support planning and project management, thereby promoting more responsive, efficient, and accountable local governance.

Development and Implementation of an Office Automation System for Barangay Local Government Units

The study by Dela Cerna and colleagues aimed to develop and implement an office automation system specifically for barangay local government units (LGUs) in the Philippines, with the objective of replacing time-consuming, paper-based administrative processes with an automated system capable of managing document issuance, record keeping, and routine transactions more efficiently. The researchers

employed the Extreme Programming (XP) agile methodology, emphasizing iterative development, continuous testing, and minimal documentation. Functional requirements were gathered through focus group discussions, preliminary investigations, and informal interviews with barangay personnel. The system was built using PHP and HTML for the front end, MySQL for the database backend, and CSS, JavaScript, Bootstrap, and Leaflet for geospatial mapping. It comprised nine core components, including residence profiling, facilities and equipment reservation, clearance and payment management, crime incidence tracking, disaster help desk, complaint filing, health care management, accounting-related functions, and a manpower skills portal. Both white-box and black-box testing were conducted to verify functionality and ensure correct input-output operations. The system was successfully deployed in Barangay Quezon on June 13, 2022, with post-deployment evaluations from ten officials, including the Barangay Captain, Secretary, Treasurer, and council members yielding perfect scores across goal attainment, program value, training effectiveness, and duration. Findings demonstrated enhanced ease of use, transparency, and automation of routine documentation tasks, significantly reducing time and resource expenditure. The study concluded that the BLGU-OAS achieved Technological Readiness Level 9 and that the XP methodology was effective in producing software closely aligned with workflow needs. Implications suggest that modular, user-centered office automation systems for barangays can substantially improve service delivery efficiency, document management accuracy, and end-user acceptance, supporting broader adoption in other local government units.

Web-based Management Information System of Cases Filed with the National Labor Relations Commission

The study conducted by Aaron Paul M. Dela Rosa of Bulacan State University aimed to describe the daily operations and challenges of the National Labor Relations Commission Regional Arbitration Branch No. IV (NLRC RAB IV) in Calamba, Laguna, and to develop a web-based management information system (MIS) to address these issues. The primary objective was to streamline the management of filed complaints, Single-Entry Approach (SEnA) cases, and labor case documents, while improving the accuracy and accessibility of case status information for Commission personnel and stakeholders. The research employed a descriptive developmental approach, gathering data through observations and interviews with NLRC staff. System development followed the Agile Software Development methodology, enabling iterative feature creation and refinement based on continuous feedback. The system was evaluated using criteria for functionality, usability, performance, and efficiency, with input

from both expert evaluators and actual end-users. Results showed overall mean evaluation scores of 4.27 from experts and 4.43 from end-users, rated as “Very Good,” indicating strong acceptance of the system. The MIS effectively managed filed complaints, SEnA cases, and labor documents, while enabling respondents and complainants to check case status online, reducing the need for in-person inquiries. The study concluded that a web-based MIS is an effective solution for government agencies handling large volumes of case documents, improving transparency, document management, and workflow efficiency. The findings imply that incorporating real-time tracking features in digital management systems enhances accountability, stakeholder satisfaction, and operational effectiveness in government processes.

Improving the Data Management System of a Local Government Unit in Baguio City, Philippines Through Information Systems Solutions: Towards Enhanced Sustainable Public Service

NThe study by Erona et al. (2025) of Saint Louis University, Baguio City, aimed to enhance the data management system of Barangay Legarda-Burnham-Kisad, which previously relied on scattered manual, paper-based records for residents and projects. The objectives were to evaluate the software quality of two developed systems. functional suitability, performance efficiency, usability, reliability, and security; to assess improvements in operational efficiency; and to determine overall enhancements in data management. The study employed a single-group pretest-posttest quantitative approach combined with action research and case study methodology. Two Excel VBA-based systems were developed: the Barangay Resident Profile Management System (BREMS) to centralize resident records and the Barangay Tracking and Recording of Accomplished Community Key-Projects (BTRACK) to structure project monitoring and reporting. Agile methodology guided iterative design sprints with continuous feedback, while Process Flow Diagrams and Data Flow Diagrams supported system analysis. System quality was evaluated using a 15-item Likert-scale questionnaire based on ISO 25010, and system effectiveness was assessed using a 12-item pretest-posttest questionnaire, with operational efficiency measured through a time study of ten simulations of resident certificate processing. All ten barangay officials participated as respondents. Results indicated significant improvements: BREMS and BTRACK achieved overall mean ratings of 4.38 and 4.80, respectively, with security scoring highest, while operational efficiency increased substantially, reducing data retrieval times from 48.21 seconds to 17.61 seconds ($p = 0.01$). Pretest-posttest effectiveness scores also rose from Satisfactory to Excellent across all categories. The study concluded that the implementation of BREMS and BTRACK successfully transformed the barangay's data management system, demonstrating that offline-capable, accessible

digital solutions can significantly improve operational efficiency, data accuracy, and reliability in local government settings, with strong implications for similar LGUs pursuing cost-effective digital transformation initiatives.

A Data-Driven System Procurement Monitoring Framework for Philippine Government Institutions

The study published in the International Journal of Engineering, Technology Research and Management (IJETRM) in December 2025 aimed to establish a structured, data-driven framework for monitoring procurement activities in Philippine government institutions, addressing inefficiencies, lack of transparency, and vulnerability to corruption in public procurement. The objective was to develop a replicable system to ensure accountability and compliance throughout the procurement cycle, from planning to contract implementation. The research employed a mixed-methods design, combining document analysis of existing procurement policies, review of procurement transaction data from selected government agencies, and consultative interviews with procurement officers and compliance auditors. The framework was developed through comparative analysis of local and global best practices, aligned with Republic Act 9184 and Government Procurement Policy Board (GPPB) guidelines, while incorporating data visualization and dashboard design principles for real-time monitoring. Results indicated that the system modules were generally well-received, with mean scores of 2.90 ("Very Acceptable") for the Admin module, 2.70 ("Acceptable" to "Very Acceptable") for the Action Personnel module, and 2.60 ("Acceptable") for the Requesting Entities module. Pilot applications demonstrated measurable improvements in procurement efficiency, faster decision-making, and enhanced detection of irregularities, with validation from the Commission on Audit (COA) confirming its alignment with audit standards. The study concluded that a data-driven procurement monitoring system is essential for fostering transparency and accountability in government institutions. The findings imply that information systems for public agencies should integrate monitoring and reporting functionalities aligned with national policy frameworks to promote compliance, generate actionable insights, and support evidence-based governance.

Local Related Literature

Digital Governance in the Philippines: A Scoping Review of Current Challenges and Opportunities (Andaya, Orlina, & Ilustre, 2025)

Andaya, Orlina, and Ilustre (2025) conducted a scoping review published in Global Sustainability Research to summarize the state of digital governance in the Philippines and examine its effects, prospects for improvement, and future opportunities. The objective was to identify achievements and persistent barriers in e-government implementation across public institutions and provide actionable recommendations for policymakers. The study employed a scoping review methodology guided by the Population, Concept, and Context (PCC) framework, using the PRISMA-ScR flow diagram to ensure transparency in study selection. A comprehensive search across Scopus, ResearchGate, and Google Scholar covering publications from 2020 to 2025 was conducted, yielding 51 initial records, of which 13 met inclusion criteria based on language, relevance, and recency. Peer-reviewed articles, government reports, and policy documents were included to capture both academic and institutional perspectives. The review found that Philippine digital governance has improved public service responsiveness, accessibility, and efficiency, particularly through e-government platforms and digitized transaction systems, but remains limited by low digital literacy among officials and citizens, insufficient ICT infrastructure in rural and isolated areas, financial constraints, and institutional resistance. Emerging technologies such as blockchain and artificial intelligence were noted as promising avenues for future research. The study concluded that while digital governance can transform public service delivery, structural and capacity challenges constrain its potential. For the current research, this implies that information systems for LGUs should prioritize inclusivity, accessibility, and functionality in low-connectivity contexts, accommodating varying levels of user digital competence.

Developing a Smart City Framework for Philippine LGUs: A Policy-Driven Approach to Digital Transformation (Bithay et al., 2025)

Bithay, Aranas, Aranas, and Armas (2025), in their study published in Edelweiss Applied Science and Technology, aimed to develop an integrative smart city framework tailored for Philippine local government units (LGUs) to address gaps in locally contextualized guidance, as existing global frameworks inadequately accounted for Philippine policy, resources, and governance structures. The study employed a mixed-methods approach, combining quantitative assessment of smart city readiness indicators benchmarked against national averages with a qualitative review of policy documents and a

case study of a pilot LGU in Central Luzon. Comparative analyses were conducted using established international frameworks including the ASEAN Smart Cities Network (ASCN) Framework, DOST Smarter City Framework, ISO 37106:2021, and the Asian Development Bank smart cities pathway model to contextualize findings. Results indicated incremental progress in areas such as digital payments, public Wi-Fi, and public safety systems, while highlighting persistent gaps in interoperability, data management standardization, and digital workforce capacity. The study concluded that a coherent, nationally aligned smart city framework is essential for guiding LGU digital transformation sustainably, emphasizing coordination between national agencies and local governments as a critical success factor. For the current research, this implies that LGU information systems should be integrated within broader digital governance ecosystems, accounting for national standards, interoperability, and institutional capacity.

Advancing E-Governance in the Philippines: Strategies for Optimizing Components and Enhancing Digital Public Service Performance (Pancho et al.)

Pancho and colleagues (2025), in their research published in Lecture Notes in Networks and Systems, examined e-governance implementation in the Philippines with the aim of identifying strategies to optimize digital public service performance and enhance citizen satisfaction. The study employed a systematic review and analysis approach, drawing on literature, government policy documents, and empirical studies across multiple Philippine government sectors, while also incorporating comparative insights from neighboring countries with successful e-governance models. Key components analyzed included ICT infrastructure, digital service delivery mechanisms, human resource development, intergovernmental coordination, and legal and regulatory frameworks. Findings indicated that improving e-governance requires addressing both supply-side factors, such as infrastructure and system design, and demand-side factors, including citizen awareness, trust, and engagement. Recommended strategies included strengthening local broadband connectivity, investing in continuous capacity-building for IT personnel, establishing interoperability standards across agencies, and developing responsive feedback mechanisms. The study concluded that meaningful e-governance improvements depend not only on technological investments but also on governance reforms, organizational change, and citizen-centric design. For the current research, this implies that information systems for LGUs should incorporate robust training, continuous monitoring, and governance support to ensure sustainable use and ongoing enhancement of digital services.

Stakeholders' Satisfaction on Service Quality of E-Governance in the Philippines (Salapa, 2025)

The study of Salapa (2025) published in the International Journal of Engineering Technology Research and Management (IJETRM), investigated stakeholder satisfaction with e-governance systems in the Philippines, focusing on how well these platforms meet user expectations regarding accessibility, responsiveness, reliability, and overall service quality. The study employed a quantitative descriptive survey methodology, administering structured questionnaires to stakeholders including government employees, business permit applicants, and ordinary citizens who interacted with e-governance systems. Satisfaction was measured across multiple service quality dimensions aligned with established e-service performance frameworks, and data were analyzed using means, standard deviations, and correlation analyses to identify significant relationships between service quality aspects and overall satisfaction. Findings indicated that while reliability and basic functionality received positive evaluations, lower satisfaction was reported for responsiveness to complaints, accessibility for users with limited digital skills, and multilingual support. Demographic factors such as age, education, and prior technology experience were found to significantly influence satisfaction levels. The study concluded that although current e-governance systems achieve basic acceptability, improvements are needed in inclusivity, user support, and responsiveness. Salapa recommended adopting user-centered design and establishing robust feedback mechanisms to enhance system performance. For the present research, this implies that government information systems should prioritize stakeholder satisfaction by integrating intuitive design, accessible features, and continuous feedback loops to ensure the system meets diverse user needs.

E-Governance: A Critical Review of E-Government Systems Features and Frameworks for Success

A literature review published on Academia.edu conducted a critical examination of existing e-government systems to identify the key features, design principles, and organizational frameworks that differentiate successful e-governance implementations from those that underperform. The study employed a critical review methodology, systematically analyzing peer-reviewed articles, government reports, and case studies on e-governance system development and evaluation. The review drew on theoretical frameworks including the Technology Acceptance Model (TAM), the DeLone and McLean Information Systems Success Model, and ISO 25000 software quality standards, incorporating both Philippine and international cases to provide a broad perspective. Findings revealed that successful e-government systems typically feature high reliability, robust data security, user-friendly interfaces, integration with organizational workflows, and strong institutional support from leadership. Systems that actively involve citizens in design and evaluation achieve higher adoption and satisfaction, whereas those lacking user consultation, training, or change management face resistance and underutilization. The review concluded that e-governance success depends not only on technical performance but also on governance, organizational alignment, and citizen engagement. For the present research, this underscores the importance of evaluating the proposed information management system not just on technical criteria, but also on its alignment with barangay processes, user acceptance, and its contribution to improved governance outcomes.

Summary

This chapter presents a comprehensive review of related literature, studies, and interrelated systems that serve as the proponents' basis for developing the Integrated Housing Services and Monitoring System for Local Government Human Settlements. The review encompasses both foreign and local sources, examining digital governance initiatives, complaint management platforms, document tracking systems, housing management solutions, and integrated public service systems.

Foreign Literature and Systems

International studies have demonstrated that digital transformation in public administration significantly enhances service delivery efficiency, transparency, and stakeholder satisfaction. The study by Szedmák et al. (2025) documented Hungary's implementation of a document management application across over 3,000 municipalities, revealing substantial reductions in administrative lead times and improved transparency. This finding directly supports the need for the proposed document management and tracking module in the IHSMS.

The research conducted by Uriawan et al. (2023) on the e-Perumahan web-based housing management system is particularly relevant to this study. It demonstrated how centralizing housing data, automating property management tasks, and improving communication among stakeholders resulted in measurable improvements in operational efficiency. The e-Perumahan platform's features including resident registration, service request submission, maintenance tracking, and complaint handling closely parallel the integrated approach proposed in the IHSMS.

Bhadouria et al. (2021) designed an online complaint management system featuring automated escalation protocols, departmental routing, and performance tracking, which successfully enhanced accountability and reduced resolution times. The system's automatic escalation feature provides a validated model for the proposed workflow and process automation module. Sierra et al. (2025) incorporated large language models into complaint management for the electric utility sector, demonstrating significant improvements in classification accuracy and response speed. Li et al. (2024) found that implementing formal complaint management systems with multichannel intake significantly improved patient satisfaction in grassroots hospitals, confirming that proper complaint categorization and tracking leads to better service outcomes.

According to David et al. (2023), successful digital technology adoption in local governments depends on strong leadership, workforce readiness, stakeholder engagement, and proper alignment with institutional goals. The researchers emphasized that organizational readiness, continuous training, and change management are equally important for sustainable digital transformation. Jiao et al. (2024) demonstrated that systematic analysis of citizen-generated data enables more refined, evidence-based governance, supporting the proposed analytics and reporting dashboard module.

Ismail (2021) examined the e-complaint method in Malaysian polytechnic institutions and identified critical weaknesses including inadequate integration with management systems and insufficient support for decision-making. The study recommended developing comprehensive systems with real-time data tracking recommendations that align with the proposed Integrated Housing Services and Monitoring System approach.

Local Literature and Systems

In the Philippine context, Andaya, Orlina, and Ilustre (2025) revealed that while digital governance has improved public service responsiveness, significant barriers persist including low digital literacy, inadequate ICT infrastructure, and institutional resistance to change. These findings emphasize the importance of designing the IHSMS with user-friendly interfaces and providing adequate training for DHSUD staff.

Pancho et al. emphasized that optimizing e-governance requires addressing both infrastructure and citizen engagement factors. This multi-pronged approach is reflected in the IHSMS design, which considers both internal office workflows and external stakeholder interactions with LGUs, developers, HOAs, and the general public.

Several Philippine studies have demonstrated successful digital transformation at the local government level. Nerona et al. (2025) developed BREMS and BTRACK for Barangay Legarda-Burnham-Kisad in

Baguio City, reducing data retrieval time from 48.21 to 17.61 seconds and improving data management quality scores from "Satisfactory" to "Excellent." This demonstrates that comprehensive information systems can dramatically improve both speed and accuracy of government operations.

Dela Cerna et al. successfully implemented the BLGU-OAS in Barangay Quezon, achieving Technological Readiness Level 9 and receiving perfect evaluation scores (5.0) from barangay officials. The system's modular design, covering nine functional components including complaint desk filing and document management, validates the proposed IHSMS's modular architecture. The study also validated the effectiveness of agile methodology, which informed the proposed IHSMS development approach.

Dela Rosa developed a web-based Management Information System for the National Labor Relations Commission that achieved "Very Good" ratings from experts (4.27) and end-users (4.43). The system's case tracking feature allowing parties to independently check case status online validates the proposed case management module's approach of providing transparency through systematic tracking from filing to resolution.

A data-driven procurement monitoring framework documented in IJETRM (2025) demonstrated that systematic monitoring mechanisms aligned with government regulations can significantly improve accountability and accelerate decision-making. This is particularly relevant as the system must support Department of Human Settlements and Ubran Development compliance monitoring functions and regulatory mandates.

Bithay et al. (2025) emphasized that information systems for Philippine LGUs should be designed as integrated components of broader digital governance ecosystems, aligned with national standards. Salapa (2025) found varying satisfaction levels with e-governance systems, with respondents expressing concerns about responsiveness and accessibility. These findings informed the IHSMS design priorities, emphasizing user-centered design and robust feedback mechanisms.

Synthesis

The findings from international and local studies establish a clear foundation for the proposed system. International research demonstrates that integrated digital platforms combining document management, complaint handling, and monitoring capabilities result in measurable improvements when properly designed. Local Philippine studies indicate that digital transformation is achievable in government environments when systems prioritize user-centered design and align with organizational workflows.

However, the reviewed literature reveals several critical gaps justifying the proposed Integrated Housing Services and Monitoring System for Local Government Human Settlements. First, most existing systems address single administrative functions in isolation, lacking comprehensive integration across workflows.

While the e-Perumahan system is comprehensive for housing management, it focuses on resident services rather than regulatory compliance monitoring. Second, there is limited evidence of integrated housing regulatory operations systems specifically designed for regional field offices like DHSUD in the Philippine context. Third, existing systems rarely connect complaints, documents, and projects within a unified framework, allowing complete contextual visibility. Fourth, many systems function as passive repositories without active deadline monitoring or workflow automation. Fifth, current systems often lack consolidated performance dashboards for evidence-based decision-making. Finally, none specifically address regulatory fee and receipt recording requirements essential to field office operations.

The proposed Integrated Housing Services and Monitoring System builds upon existing principles while addressing gaps with key innovations such as unified integration of documents, complaints, projects, workflow automation, and fee tracking; relational linking across record types; active deadline monitoring with escalation triggers; regulatory fee recording and analytics-driven dashboards supporting evidence-based management.

By combining insights obtained from successful implementations in Hungary's municipalities, Malaysia's polytechnic institutions, Philippine barangays, and Philippine national agencies, the proponents propose the Integrated Housing Services and Monitoring System as a solution for housing regulatory field offices. This study directly discusses the gaps highlighted in Department of Human Settlements and Urban devlopment-NIR Bacolod Office operations that include difficulty tracking documents, unstructured complaint processing, disconnected monitoring and compliance, slow manual coordination, and inefficient fee recording.

References:

David, Anne, et al. "Understanding Local Government Digital Technology Adoption Strategies: A PRISMA Review." *Sustainability*, vol. 15, no. 12, 1 Jan. 2023, p. 9645, www.mdpi.com/2071-1050/15/12/9645, https://doi.org/10.3390/su15129645

Abuimara, T., Hobson, B. W., Gunay, B., O'Brien, W., & Kane, M. (2021). Current state and future challenges in building management: Practitioner interviews and a literature review. *Journal of Building Engineering, 41,* 102803. https://www.sciencedirect.com/science/article/abs/pii/S2352710221006616

Alshaikh, M., & El-Asmar, M. (2023). A systematic review of construction project performance metrics. *Canadian Journal of Civil Engineering, 51*(1), 3–18. https://journals.sagepub.com/doi/full/10.1177/08445621231203086

Ismail, Z.-A. (2021, October 13). *Exploring e-complaint method: Learning from the Malaysian Polytechnic institutions*. Technical paper. https://www.emerald.com/jfm/article-abstract/20/3/501/224306/Exploring-e-complaint-method-learning-from-the?redirectedFrom=fulltext

Jiao, Y., Li, C., Yao, Z., Weng, C., Lian, A., & Dong, R. (2024). How can online citizen complaints provide solutions to refine environmental management: A spatio-temporal perspective. *Information Processing & Management, 61*(2), 103611. https://www.sciencedirect.com/science/article/abs/pii/S0306457323003485

Szedmák, Borbála, et al. "Digital Transformation of Public Services: The Case of the Document Management Application." *International Journal of Public Administration*, 3 July 2025, pp. 1–18, https://doi.org/10.1080/01900692.2025.2520522.

Bhadouria, L. S., Kumar, N., Faisal, A., & Devi, S. (2021). "Online complaint management system." Gurgaon, 12(6), 5144–5150. https://www.proquest.com/openview/41ae97c3a4e85df27ba10de9797c7577/1?pq-origsite=gscholar&cbl=2045096

Uriawan, W., Noorsyahbannie, M. N. B., Lestari, I. S., Solehudin, M. A., & Enjelianto, I. (2023, July 19). *Revolutionizing property management: e-Perumahan web-based housing management system*. https://www.preprints.org/frontend/manuscript/40d64e74563bf24ca361b7d1e7d85e31/download_pub

Sierra, C., Huanca, F., Torres, M., & Núñez, F. (2025). An integrated complaint management system based on large language models: Case study in the electric sector. *Array, 28,* 100570. https://www.sciencedirect.com/science/article/pii/S2590005625001973

Li, G., Chen, Y., & Lou, X. (2024). Complaint management system and patient satisfaction in grassroots hospitals. *Medicine, 103*(8), e37275. https://journals.lww.com/md-journal/fulltext/2024/02230/complaint_management_system_and_patient.38.aspx?context=latestarticles

Andaya, Eduardo, et al. "Digital Governance in the Philippines: A Scoping Review of Current Challenges and Opportunities." Ssrn.com, 18 Jan. 2025, papers.ssrn.com/sol3/papers.cfm?abstract_id=5198618.\

Bithay, Leovigilda L, et al. "Developing a Smart City Framework for Philippine LGUs: A Policy-Driven Approach to Digital Transformation." Edelweiss Applied Science and Technology, vol. 9, no. 7, 21 July 2025, pp. 1704–1719, www.researchgate.net/publication/394701712

Pancho, Ronald L, et al. "Advancing E-Governance in the Philippines: Strategies for Optimizing Components and Enhancing Digital Public Service Performance." Lecture Notes in Networks and Systems, https://doi.org/10.1007/978-3-032-12993-2_8.

Salapa, Aristeo. "STAKEHOLDERS' SATISFACTION on SERVICE QUALITY of E-GOVERNANCE in the PHILIPPINES." International Journal of Engineering Technology Research & Management, vol. 12, 2025, ijetrm.com/issues/files/Dec-2025-08-1765199923-DEC23.pdf.

e-Governance: A Critical Review of e-Government Systems Features and Frameworks for Success https://www.academia.edu/download/108156711/414-1-614-1-10-20230420.pdf

Improving the data management system of a local government unit in Baguio City, Philippines through information systems solutions: Towards enhanced sustainable public service. Journal of Intelligent Systems and Applied Data Science (JISADS),Vol.3, Issue.2, (2025) PP. 72-84https://jisads.com/index.php/1/article/view/46/46

Information management system with project monitoring for barangay. (2023). Journal of Engineering and Applied Innovations, Vol.3(Series 2023), 51–60. http://urdc.usl.edu.ph/journals/jeai/papers/vol3/vol%203%20series%202023-51-60.pdf

Dela Cerna, M. (2023). Development and implementation of an office automation system for barangay local government units. ResearchGate.

https://www.researchgate.net/profile/Monalee-Dela-Cerna-2/publication/.pdf

Dela Rosa, A. P. M. (2022). Web-based management information system of cases filed with the National Labor Relations Commission. International Journal of Computing Sciences Research. Advance online publication. doi:10.25147/ijcsr.2017.001.1.113 arXiv.https://arxiv.org/pdf/2211.14452

A data-driven system procurement monitoring framework for Philippine government institutions. (2025). International Journal of Engineering Technology Research & Management. Volume-09 Issue 12. https://ijetrm.com/issues/files/Dec-2025-15-1765822414-DEC38.pdf